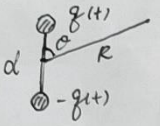
上节回顾

电偶极辐射



振动的电偶极子

$q(t) = q_0 \cos\omega t$

$p(t) = p_0 \cos\omega t \qquad p_0 = q_0 d$

$d \ll \lambda, R$

$$\varphi(\vec{x}, t) = \frac{p_0 \cos\theta}{4\pi\varepsilon_0 R}\left\{-k\sin[\omega(t-\tfrac{R}{c})] + \frac{1}{R}\cos[\omega(t-\tfrac{R}{c})]\right\}$$

远场辐射区 $R \gg \lambda$

$$\varphi(\vec{x}, t) \approx -\frac{p_0\omega}{4\pi\varepsilon_0 c}\left(\frac{\cos\theta}{R}\right)\sin[\omega(t-\tfrac{R}{c})] = \frac{\dot{\vec{p}}(t-\frac{R}{c})\cdot\vec{e}_R}{4\pi\varepsilon_0 cR}$$

$$\vec{A}(\vec{x}, t) \approx \frac{-\mu_0 p_0 \omega}{4\pi R}\sin[\omega(t-\tfrac{R}{c})]\vec{e}_z = \frac{\mu_0\dot{\vec{p}}(t-\frac{R}{c})}{4\pi R}$$

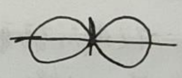
$$\vec{B} = \nabla\times\vec{A} = \frac{-\mu_0 p_0\omega^2}{4\pi c}\left(\frac{\sin\theta}{R}\right)\cos[\omega(t-\tfrac{R}{c})]\vec{e}_\phi$$

$$= \frac{\mu_0\omega^2}{4\pi cR}\vec{e}_R\times\vec{p}(t-\tfrac{R}{c})$$

$$\vec{E} = -\nabla\varphi - \frac{\partial\vec{A}}{\partial t} = -\frac{\mu_0 p_0\omega^2}{4\pi}\frac{\sin\theta}{R}\cos[\omega(t-\tfrac{R}{c})]\vec{e}_\theta$$

$$= -\frac{\mu_0\omega^2}{4\pi R}\vec{e}_R\times(\vec{e}_R\times\vec{p}(t-\tfrac{R}{c}))$$

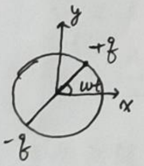
$$\vec{S} = \frac{1}{\mu_0}(\vec{E}\times\vec{B}) = \frac{\mu_0}{c}\left\{\frac{p_0\omega^2}{4\pi}\frac{\sin\theta}{R}\cos[\omega(t-\tfrac{R}{c})]\right\}^2\vec{e}_R$$



$$\langle\vec{S}\rangle = \left(\frac{\mu_0 p_0^2\omega^4}{32\pi^2 c}\right)\frac{\sin^2\theta}{R^2}\vec{e}_R \qquad \propto \frac{1}{R^2},\ p_0^2,\ \omega^4$$

$I_0 = q_0\omega \qquad \lambda = \frac{2\pi c}{\omega}$

总辐射功率 $\langle P_{电}\rangle = \int\langle\vec{S}\rangle\cdot d\vec{a} = \frac{\mu_0 p_0^2\omega^4}{12\pi c} = \frac{\mu_0 q_0^2 d^2\omega^4}{12\pi c} = \frac{\pi}{3}\sqrt{\frac{\mu_0}{\varepsilon_0}}I_0^2\left(\frac{d}{\lambda}\right)^2$



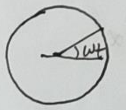
转动的电偶极子（作业）

$\vec{p} = p_0[\cos(\omega t)\vec{e}_x + \sin(\omega t)\vec{e}_y]$

复数形式 $\vec{p} = p_0 e^{-i\omega t}(\vec{e}_x + i\vec{e}_y)$

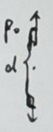
球坐标系下. $\vec{p} = p_0 e^{-i\omega t + i\varphi}[\sin\theta\,\vec{e}_R + \cos\theta\,\vec{e}_\theta + i\vec{e}_\varphi]$

$$\begin{pmatrix}\vec{e}_x\\ \vec{e}_y\\ \vec{e}_z\end{pmatrix} = \begin{pmatrix}\sin\theta\cos\varphi & \cos\theta\cos\varphi\ \cancel{\sin\theta\sin\varphi} & -\sin\varphi\ \cancel{\cos\varphi} \\ \sin\theta\sin\varphi & \cos\theta\sin\varphi & \cos\varphi \\ \cos\theta & -\sin\theta & 0\end{pmatrix}\begin{pmatrix}\vec{e}_R\\ \vec{e}_\theta\\ \vec{e}_\varphi\end{pmatrix}$$



带电粒子做圆周运动（电子绕原子核运动）

$\vec{p} = ea(\cos\omega t\,\vec{e}_x + \sin\omega t\,\vec{e}_y)$



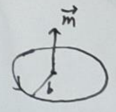
电四极辐射模型.（两个相反方向振动的电偶极子）

可利用振动的电偶极子的结果 + 叠加原理 加以分析（作业）

$$\vec{A}_{e4} = -\frac{\mu_0 p_0\omega^2 d}{4\pi cR}\cos\theta\cos[\omega(t-\tfrac{R}{c})]\vec{e}_z \qquad \cos[\omega(t-\tfrac{R}{c})]$$

$$= \frac{\mu_0}{24\pi Rc}\dddot{\vec{D}} \qquad \vec{D} = \overleftrightarrow{D}\cdot\vec{e}_R \qquad \overleftrightarrow{D} = 6p_0 d\,\vec{e}_z\vec{e}_z$$

磁偶极辐射



$I = I_0\cos\omega t$

$\vec{m} = I_0\pi b^2\cos\omega t\,\vec{e}_z = m_0\cos\omega t\,\vec{e}_z$

$P = \frac{4\pi^3}{15}\sqrt{\frac{\mu_0}{\varepsilon_0}}\left(\frac{d}{\lambda}\right)^4 I_0^2$

$b \ll \lambda, R$

$$\vec{A} \approx \frac{\mu_0 m_0}{4\pi}\frac{\sin\theta}{R}\left\{\frac{1}{R}\cos[\omega(t-\tfrac{R}{c})] - \frac{\omega}{c}\sin[\omega(t-\tfrac{R}{c})]\right\}\vec{e}_\phi$$

远场辐射区 $R \gg \lambda$

$$\vec{A} \approx -\frac{\mu_0 m_0 \omega}{4\pi c}\frac{\sin\theta}{R}\sin[\omega(t-\tfrac{R}{c})]\vec{e}_\phi = -\frac{\mu_0}{4\pi R c}\vec{e}_R \times \dot{\vec{m}}(t-\tfrac{R}{c})$$

$$\vec{B} = \nabla\times\vec{A} = -\frac{\mu_0 m_0 \omega^2}{4\pi c^2}\left(\frac{\sin\theta}{R}\right)\cos[\omega(t-\tfrac{R}{c})]\vec{e}_\theta = -\frac{\mu_0 \omega^2}{4\pi R c^2}\vec{e}_R\times[\vec{e}_R\times(\vec{m}(t-\tfrac{R}{c})]$$

$$\vec{E} = -\frac{\partial \vec{A}}{\partial t} = \frac{\mu_0 m_0 \omega^2}{4\pi c}\frac{\sin\theta}{R}\cos[\omega(t-\tfrac{R}{c})]\vec{e}_\varphi$$

[$\vec{p}\to\frac{\vec{m}}{c}$, $E\to c\vec{B}$. $c\vec{B}\to-\vec{E}$, 即可由电偶极子的场推出磁偶极子的场]

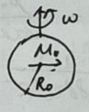
$$\vec{S} = \frac{1}{\mu_0}(\vec{E}\times\vec{B}) = \frac{\mu_0}{c}\left\{\frac{m_0\omega^2}{4\pi c}\frac{\sin\theta}{R}\cos[\omega(t-\tfrac{R}{c})]\right\}^2\vec{e}_R$$

$$\langle \vec{S}\rangle = \frac{\mu_0 m_0^2 \omega^4}{32\pi^2 c^3}\frac{\sin^2\theta}{R^2}\vec{e}_R$$

$$\langle P_{磁}\rangle = \frac{\mu_0 m_0^2\omega^4}{12\pi c^3} = \frac{\mu_0(\pi b^2 I_0)^2\omega^4}{12\pi c^3} = \frac{4\pi^5}{3}\sqrt{\frac{\mu_0}{\varepsilon_0}}I_0^2\left(\frac{b}{\lambda}\right)^4$$

$$\frac{\langle P_{磁}\rangle}{\langle P_{电}\rangle} = 4\pi^4\left(\frac{b}{\lambda}\right)^4\left(\frac{\lambda}{d}\right)^2 = \cancel{(kb)^2}(kb)^2 \ll 1 \qquad (d=\pi b)$$

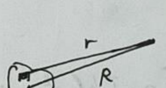
转动的均匀永磁体小球 (转动的磁偶极子)



$m_0 = M_0\cdot\frac{4}{3}\pi R_0^3$

$$\vec{m} = m_0(\cos\omega t\,\vec{e}_x + \sin\omega t\,\vec{e}_y)$$

任意源的辐射 (矢势的多极展开)



$$f(r) = f(R) - \vec{x}'\cdot\nabla f(R) + \cdots$$

$$\vec{A}(\vec{x},t) = \frac{\mu_0}{4\pi}\int\frac{\vec{J}(\vec{x}',t-\frac{r}{c})}{r}dV'$$

$$= \frac{\mu_0}{4\pi}\int\frac{\vec{J}(\vec{x}',t-\frac{R}{c})}{R}dV' - \frac{\mu_0}{4\pi}\int\vec{x}'\cdot\nabla\frac{\vec{J}(\vec{x}',t-R/c)}{R}dV' + \cdots$$

$$\int\vec{J}\,dV' = \int\rho\vec{v}\,dV' = \frac{d}{dt}\int\rho\vec{x}'\,dV' = \frac{d}{dt}\vec{p}$$

$\vec{A}^{(1)} = \frac{\mu_0}{4\pi R}\dot{\vec{p}}(t-\frac{R}{c})$ [振荡的电偶极矩产生的势]

$$\vec{A}^{(2)} = -\frac{\mu_0}{4\pi}\int\vec{x}'\cdot\Big[\underbrace{\frac{\nabla\vec{J}(\vec{x}',t-R/c)}{R}}_{\propto\frac{k}{R}} + \underbrace{\vec{J}(\vec{x}',t-R/c)\nabla\frac{1}{R}}_{\propto\frac{1}{R^2}}\Big]dV'$$

考虑远场 ($\cancel{kR}$ $R\gg\lambda$) 及单频情况 ($\nabla\to i\vec{k}$), 可忽略第二项

$$\vec{A}^{(2)} \approx -\frac{i\mu_0 k}{4\pi R}\int\vec{e}_R\cdot\vec{x}'\vec{J}(\vec{x}',t-R/c)\,dV'$$

$$= -i\frac{\mu_0 k}{4\pi R}\frac{1}{2}\vec{e}_R\cdot\left[\int(\vec{x}'\vec{J}+\vec{J}\vec{x}')dV' + \int(\vec{x}'\vec{J}-\vec{J}\vec{x}')dV'\right]$$

先看第2项

$$(\vec{e}_R\cdot\vec{x}')\vec{J}-(\vec{e}_R\cdot\vec{J})\vec{x}' = -\vec{e}_R\times(\vec{x}'\times\vec{J})$$

$$-\vec{e}_R\times\int\frac{1}{2}\vec{x}'\times\vec{J}\,dV' = -\vec{e}_R\times\vec{m}$$

$$\vec{A}_m^{(2)} = \frac{i\mu_0\omega}{4\pi Rc}\vec{e}_R\times\vec{m}(t-\frac{R}{c}) = -\frac{\mu_0}{4\pi Rc}\dot{\vec{m}}(t-\frac{R}{c})$$ [磁偶极子的贡献]

再看第一项

$$\frac{1}{2}\int[(\vec{e}_R\cdot\vec{x}')\vec{J}+(\vec{e}_R\cdot\vec{J})\vec{x}']\,dV'$$

$$=\frac{1}{2}\int[(\vec{e}_R\cdot\vec{x}')\rho\vec{v}+(\vec{e}_R\cdot\rho\vec{v})\vec{x}']\,dV'$$

$$=\frac{1}{2}\frac{d}{dt}\int(\vec{e}_R\cdot\vec{x}')\vec{x}'\rho\,dV'$$

$$=\frac{1}{6}\vec{e}_R\cdot\dot{\overleftrightarrow{D}} \qquad \overleftrightarrow{D}=\int 3\rho\vec{x}'\vec{x}'\,dV'$$

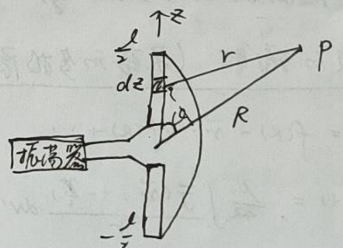
$$\vec{A}_e^{(2)} = \frac{-i\mu_0\omega}{4\pi Rc}\frac{1}{6}\underbrace{\vec{e}_R\cdot\dot{\overleftrightarrow{D}}}_{\dot{\vec{D}}} = \frac{-i\mu_0\omega}{24\pi Rc}\dot{\vec{D}} = \frac{\mu_0}{24\pi Rc}\ddot{\vec{D}}$$ [电四极矩的贡献]

$$\vec{B}_e^{(2)} = \nabla\times\vec{A}_e^{(2)} = ik\vec{e}_R\times\vec{A}_e^{(2)} = \frac{\mu_0}{24\pi Rc^2}\dddot{\vec{D}}\times\vec{e}_R$$

$$\vec{E}_e^{(2)} = c\vec{B}_e^{(2)}\times\vec{e}_R = \frac{\mu_0}{24\pi Rc}(\dddot{\vec{D}}\times\vec{e}_R)\times\vec{e}_R$$

$$\langle\vec{S}\rangle = \frac{1}{4\pi\varepsilon_0}\frac{1}{288\pi c^5R^2}(\dddot{\vec{D}}\times\vec{e}_R)^2\vec{e}_R$$

线型天线辐射.



$$I(z',t) = I_0e^{-i\omega t}\sin[k(\frac{l}{2}-|z'|)]$$

$$A_z = \frac{\mu_0I_0}{4\pi}\int_{-\frac{l}{2}}^{\frac{l}{2}}\frac{e^{-i\omega(t-\frac{r}{c})}\sin[k(\frac{l}{2}-|z'|)]}{r}\,dz'$$

远场条件下 (分母 $r\approx R$, 分子相因子中 $r\approx R-z'\cos\theta$)

$$A_z \approx \frac{\mu_0I}{4\pi R}e^{-i\omega(t-\frac{R}{c})}\int_{-\frac{l}{2}}^{\frac{l}{2}}e^{-ikz'\cos\theta}\sin[k(\frac{l}{2}-|z'|)]\,dz'$$

$$=\frac{\mu_0I}{2\pi Rk}\frac{\cos(\frac{kl}{2}\cos\theta)-\cos(\frac{kl}{2})}{\sin^2\theta}e^{-i\omega(t-\frac{R}{c})}$$

$$\vec{B} = \nabla\times\vec{A} = ik\vec{e}_R\times\vec{A} = -ikA_z\sin\theta\,\vec{e}_\varphi$$

$$\vec{E} = c\vec{B}\times\vec{e}_R = -ickA_z\sin\theta\,\vec{e}_\theta$$

$$\langle\vec{S}\rangle = \frac{1}{2\mu_0}\mathrm{Re}(\vec{E}\times\vec{B}^*) = \frac{\mu_0I_0^2c}{8\pi^2R^2}\left[\frac{\cos(\frac{kl}{2}\cos\theta)-\cos\frac{kl}{2}}{\sin\theta}\right]^2\vec{e}_R = f(\theta,\varphi)R^2\vec{e}_R$$

讨论: 1) 当 $l\ll\lambda$, 即 $kl\ll1$ 时

$$f(\theta,\varphi)\approx\frac{\mu_0I_0^2c}{8\pi^2}\left[\frac{\frac{1}{2}(\frac{kl}{2}\cos\theta)^2-\frac{1}{2}(\frac{kl}{2})^2}{\sin\theta}\right]^2 = \frac{\mu_0\omega^4}{32\pi^2c}\sin^2\theta\left(\frac{I_0l^2}{4c}\right)^2$$

对比电偶极子的辐射角分布可知，短波天线等价于一个电偶极子 $P_0=\frac{Z_0 l^2}{4c}$

对短天线，$l\uparrow$，$\langle S\rangle\uparrow$

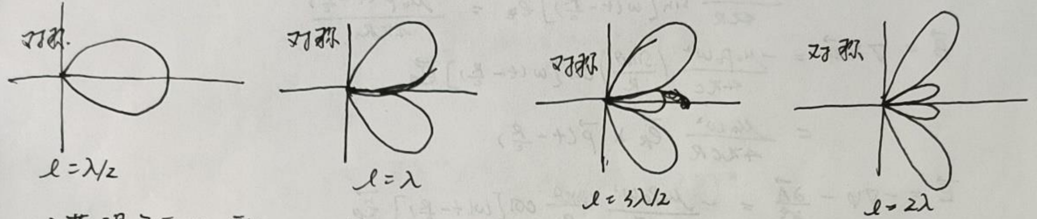
2) 当 $\frac{l}{\lambda}=\frac{1}{2},\frac{3}{2},\cdots$ 时 $\langle S\rangle$ 达到极值，$l=\frac{\lambda}{2}$ 时 $\langle S\rangle$ 最大。

总辐射功率（$l=\frac{\lambda}{2}$ 时） $P=2.44\frac{\mu_0 c I_0^2}{8\pi}\approx 36.56\,I_0^2$ （计算过程见书上）

辐射电阻 $R_r=\frac{2P}{I_0^2}\approx 73.13\,\Omega$

3) 辐射的方向性。

随着 $\frac{l}{\lambda}\uparrow$，辐射图案与偶极子辐射图案有明显的区别，开始有分叉出现，并开始朝天线方向集中。

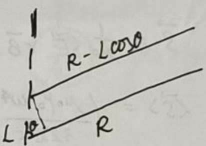


为了获得方向性更好的辐射，使用天线阵。

天线阵

设第一个天线的辐射场为

$$\vec{E}_1=\vec{C}(\theta)\frac{e^{ikR_1}}{R_1}$$



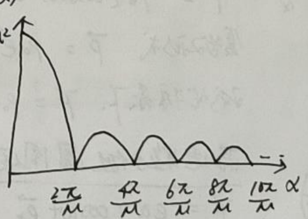
则

$$\vec{E}_2=\vec{C}(\theta)\frac{e^{ikR_2}}{R_2}\approx\vec{E}_1 e^{ikL\cos\theta}=\vec{E}_1 e^{i\alpha}$$

$$\vec{E}=\sum_{n=0}^{M-1}\vec{E}_1 e^{in\alpha}=\vec{E}_1\frac{1-e^{iM\alpha}}{1-e^{i\alpha}}$$

辐射角分布多个因子

$$f(\alpha)=\left|\frac{1-e^{iMkL\cos\theta}}{1-e^{ikL\cos\theta}}\right|^2=\frac{\sin^2(\frac{M}{2}kL\cos\theta)}{\sin^2(\frac{1}{2}kL\cos\theta)}=\frac{\sin^2(M\frac{\alpha}{2})}{\sin^2(\frac{\alpha}{2})}$$



讨论. 1) 当 $\alpha=0$ 时 $f(\alpha)$ 有极大值 $f(0)=M^2$

说明向前传播的方向辐射最强，而且辐射增强了 M^2 倍。

M 越大，主辐瓣的张角越小，方向性越强

$\sin\psi=\cos\theta=\frac{\lambda}{ML}$ $\quad(kL\cos\theta=\frac{2\pi}{M})$